Recent Advances in Root Detection with GPR

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Why the interest in tree roots?

- Effect of silvicultural or horticultural practices
- Carbon sequestration and accounting
- Tree health and protection
- Soil stability
- Hydrology

Why use GPR to assess tree roots?

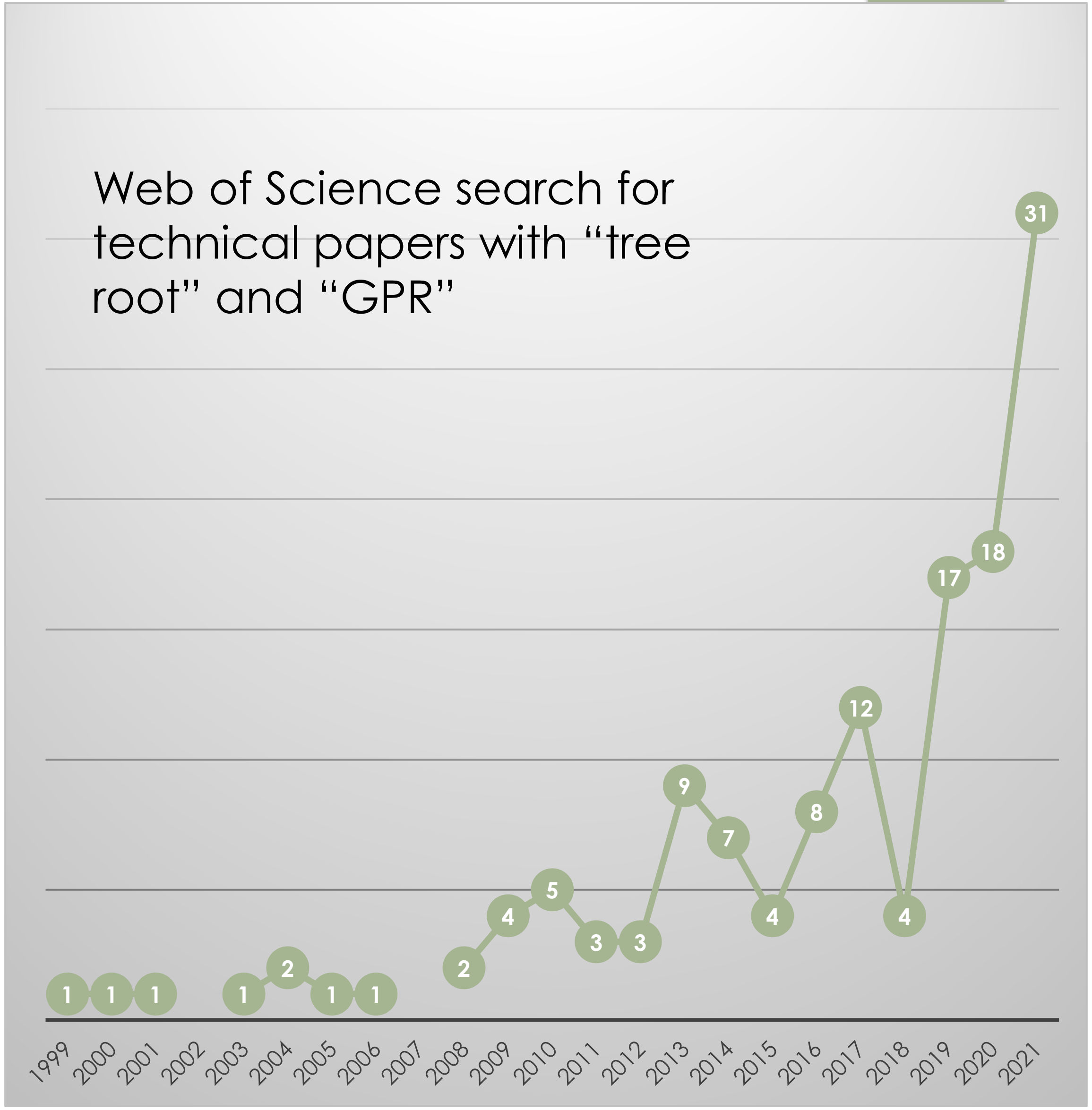
- **Non-destructive**
- Repeatable
- Ability to assess more land area
- Time and labor
- Cost

Interest in detecting roots with GPR

- Interest has grown tremendously over the last 20 years
- Expanded access to equipment
- Antenna, data storage and collection equipment have improved

Limitations to wider adoption?

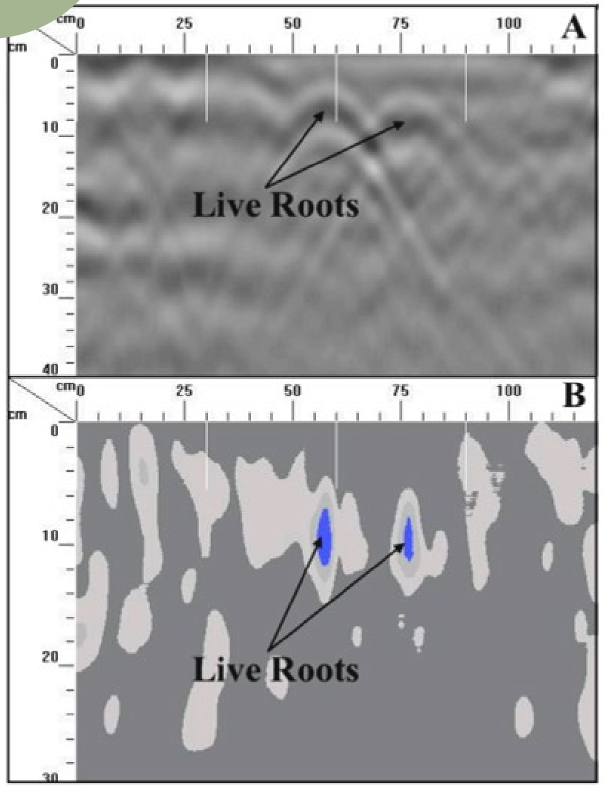
- Challenging environmental conditions for data collection
- Post-collection processing
- Few options for non-experts



My first GPR/root paper

What root and rhizosphere properties can be assessed?

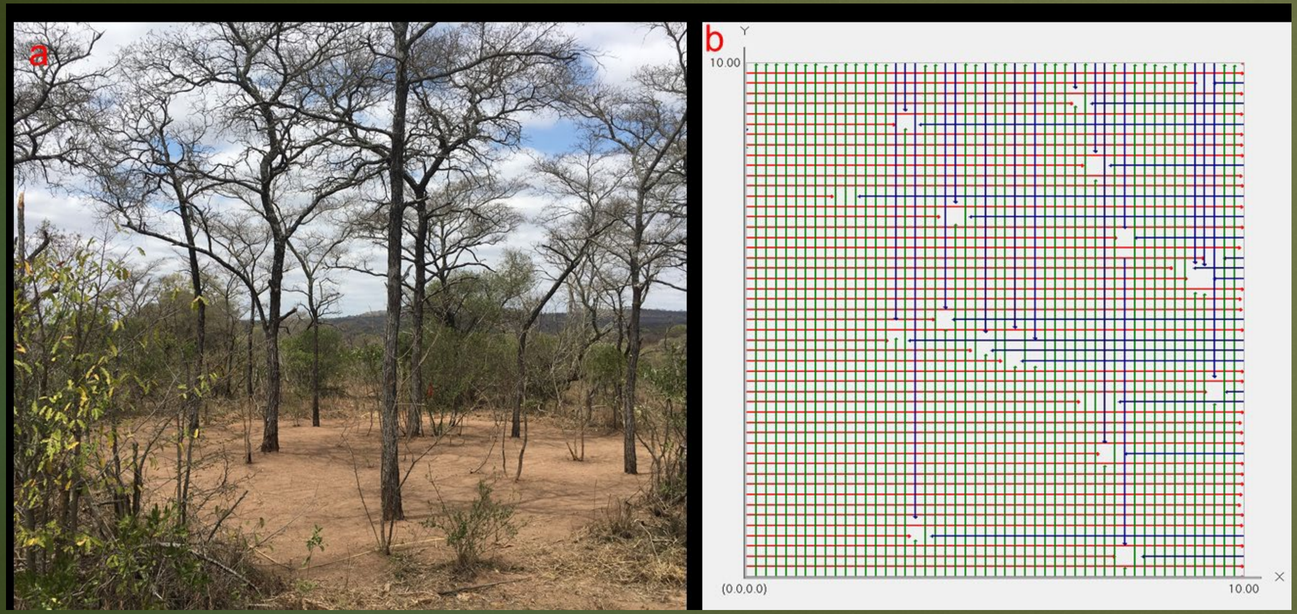
- Biomass
- Spatial distribution
- Depth
- Diameter
- Root length
- Root volume
- Rooting zone
- Phenotyping
- Architecture
- Soil moisture
- Root moisture
- Seasonal or permafrost



Bain *et al.* 2017. Remote Sensing.

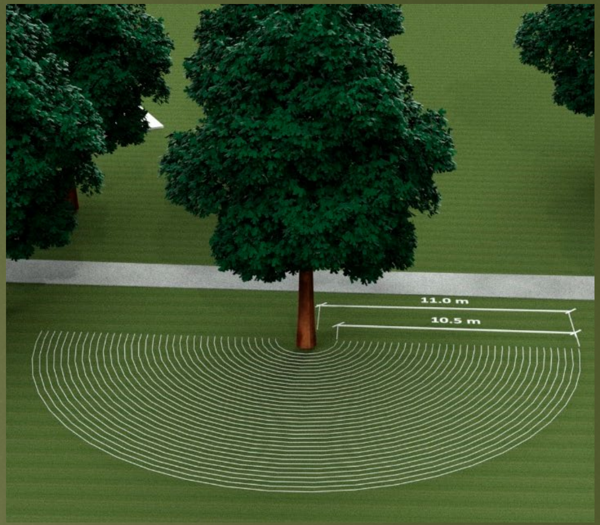
Data collection

Parallel transects or grids



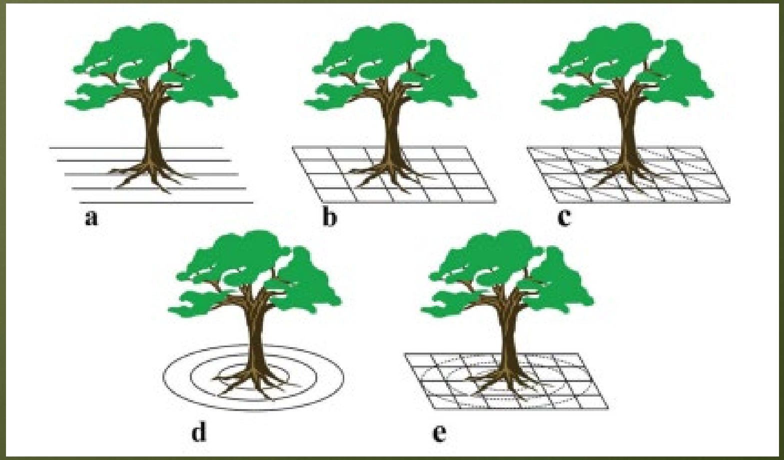
Zhou *et al.* 2022. Nature.

Polar plot



Lantini *et al.* 2022. Remote Sensing.

Hybrid approach

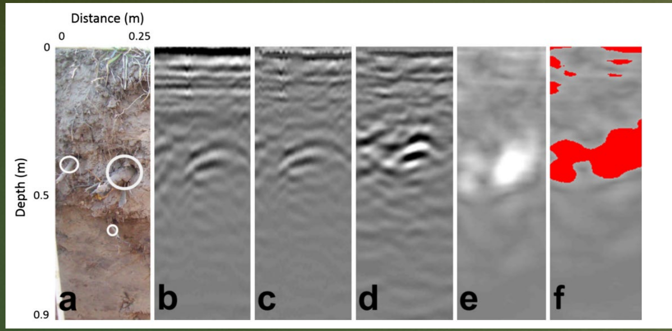


Guo *et al.* 2015. Plant and Soil.

Approaches to root analysis

Image analysis

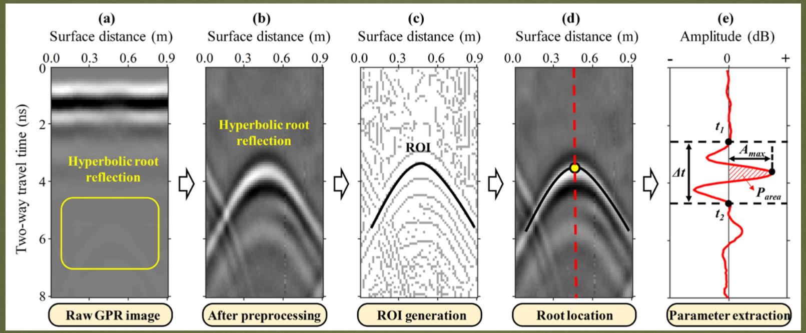
- Quantitative data



Borden *et al.* 2014. Agroforestry Systems.

Waveform analysis

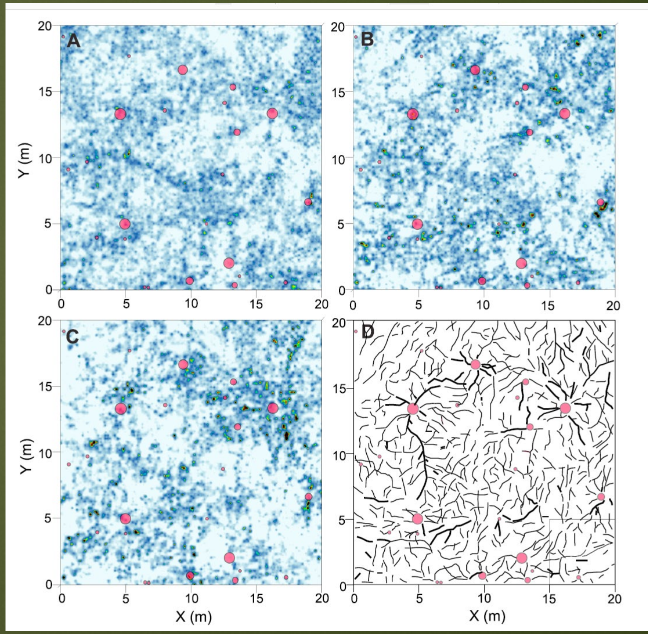
- High resolution quantitative data



Liu *et al.* 2022. IEEE TGRS.

3D radar

- Spatial mapping and visual data

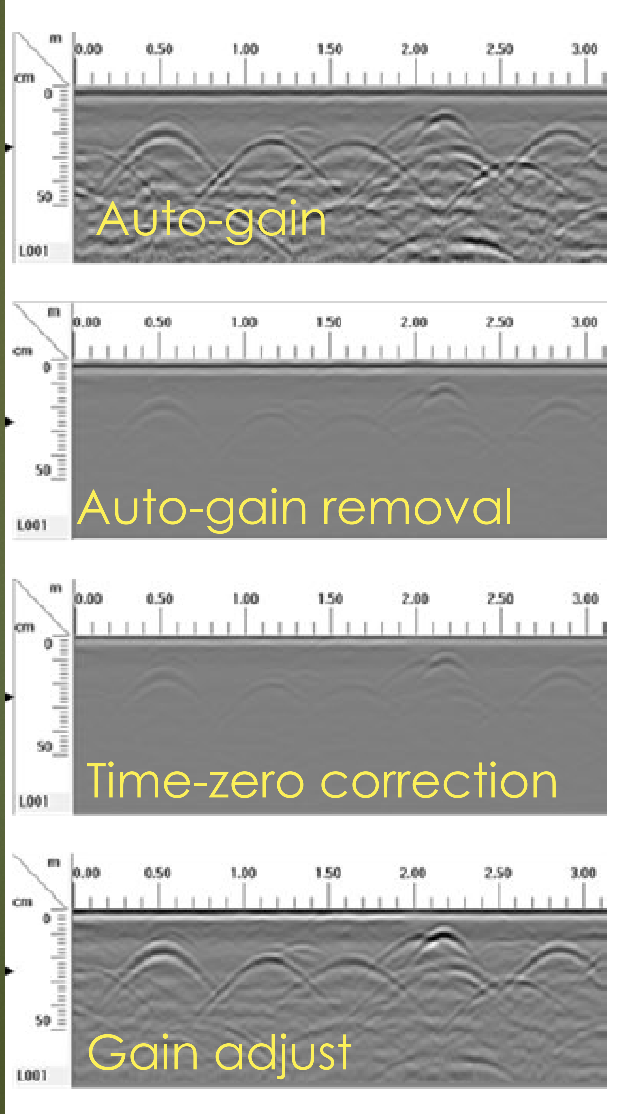


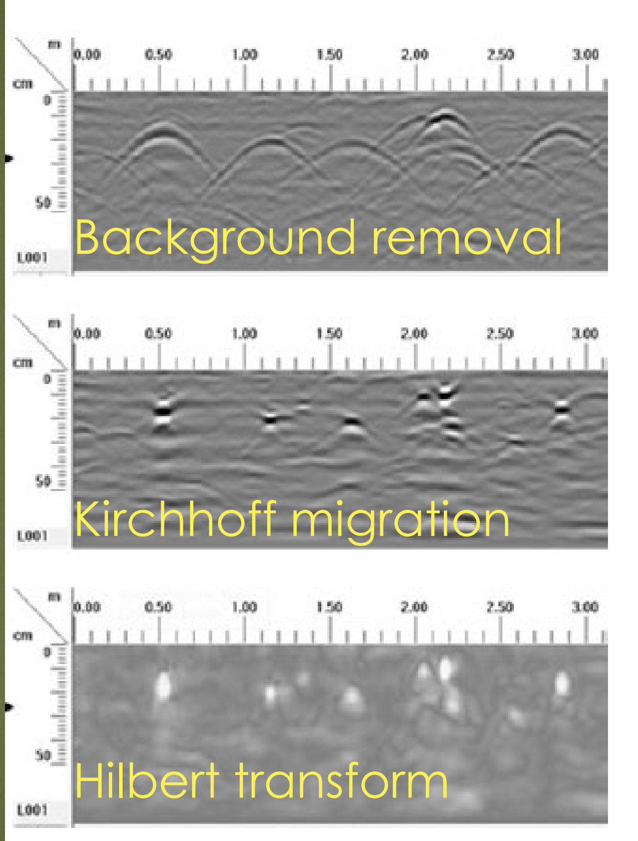
Molon *et al.* 2016. Biogeosciences

Approaches to root analysis

Image analysis

- Process radargrams to compact reflections to the size of the reflector
- Use regression to relate threshold area to biomass from cores or pits
- Use image analysis techniques to sub-sample extensive transect
- Quantitative data





Zhou *et al.* 2022. Nature.

Approaches to root analysis

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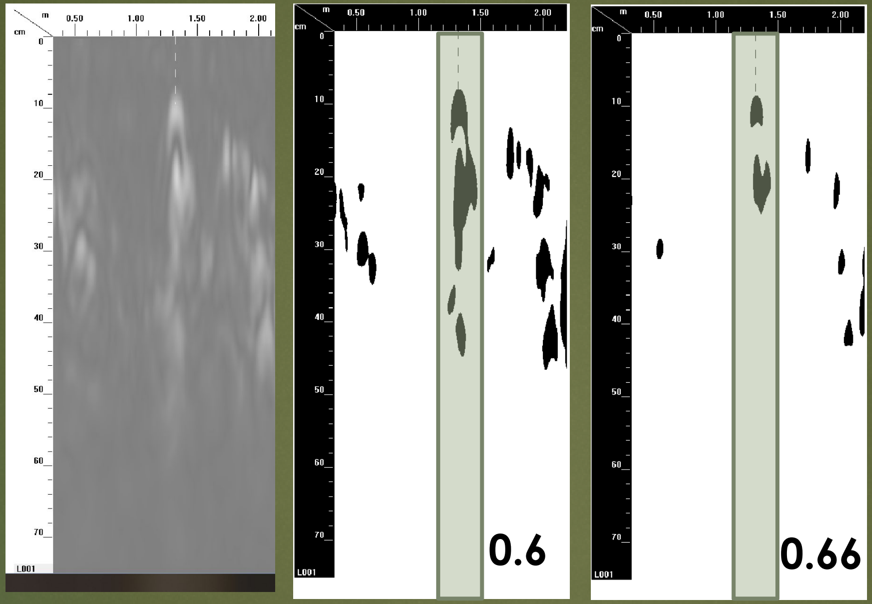
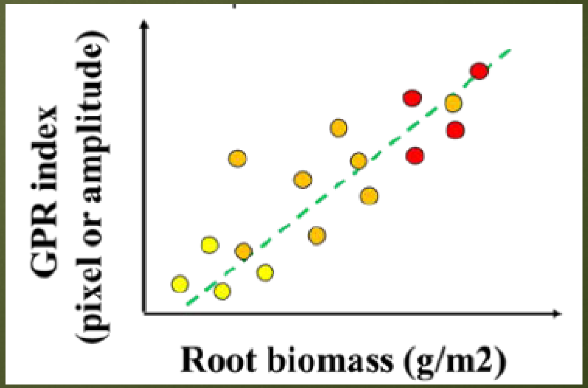


Intensity Threshold

Validation



Regression

Zhou *et al.* 2022. Nature.

Recent advances in image analysis

GPRimage

- R package (in development) to streamline data analysis
- File conversion tool
- Image viewer
- Soil core autoanalyzer
- Transect autoanalyzer with gridding functions

Authors:

John Butnor, USDA Forest Service

Yong Zhou, Yale University, starting at Utah State University in the September.

Dependent Packages

RGPR

http://emanuelhuber.github.io/RGPR/

EBImage

https://www.bioconductor.org/packages/release/bioc/html/EBImage.html

Shinyapps

https://www.shinyapps.io/

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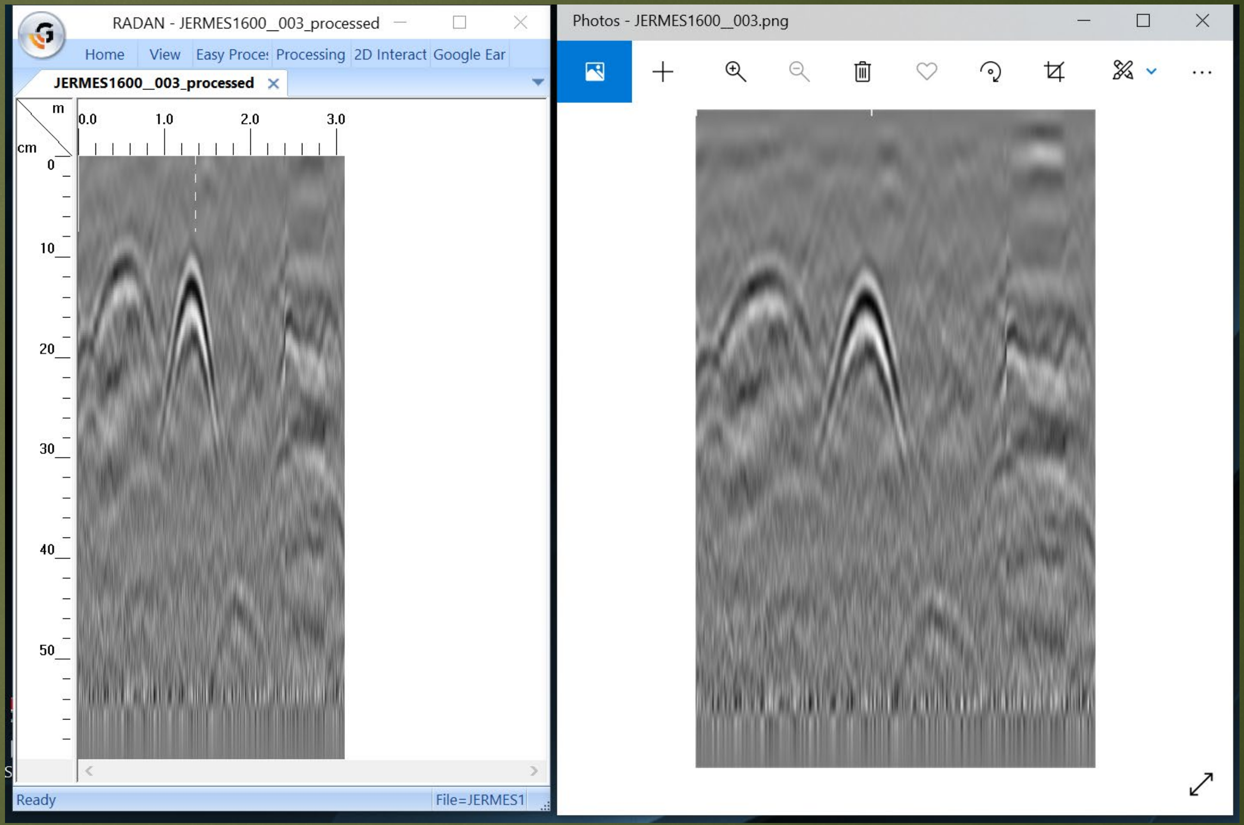

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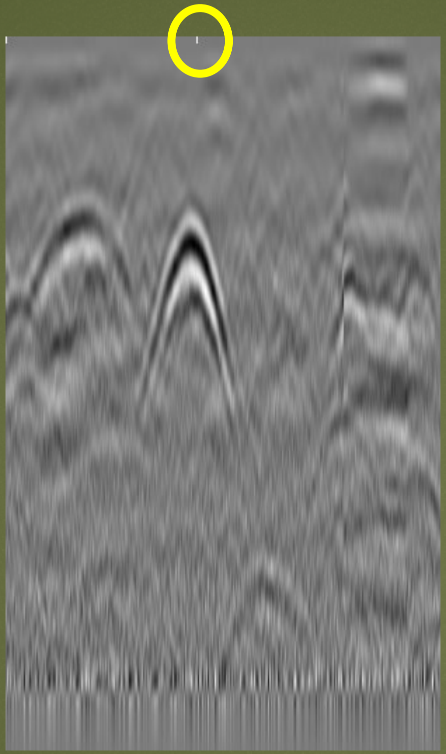
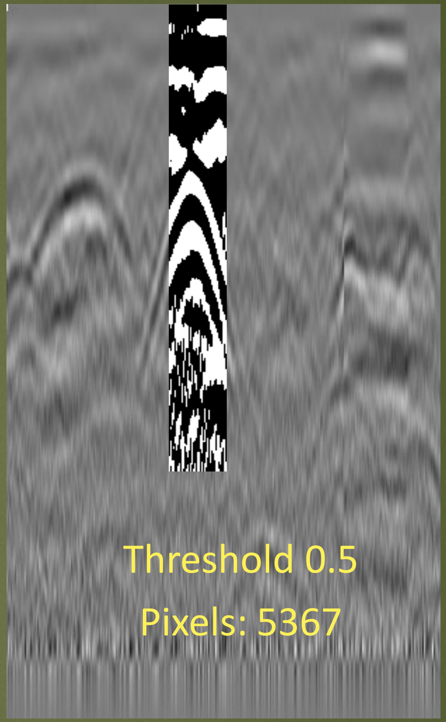
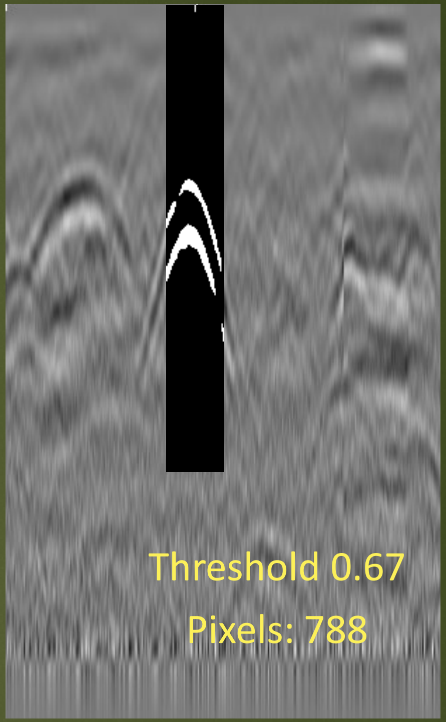


Autodetects user marks





Subsets ROI and exports multiple threshold values for comparison with calibration cores

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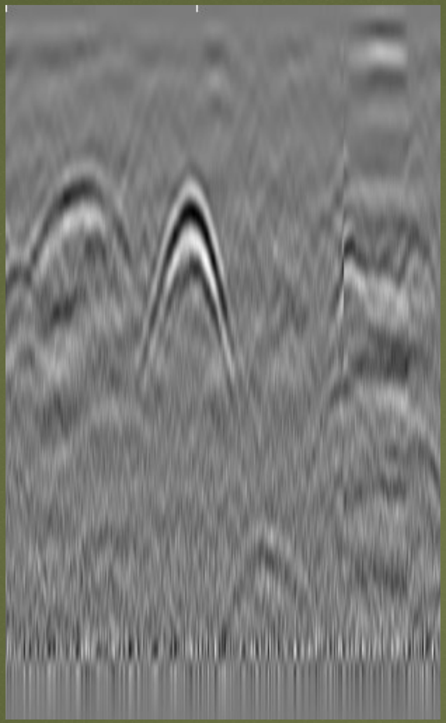

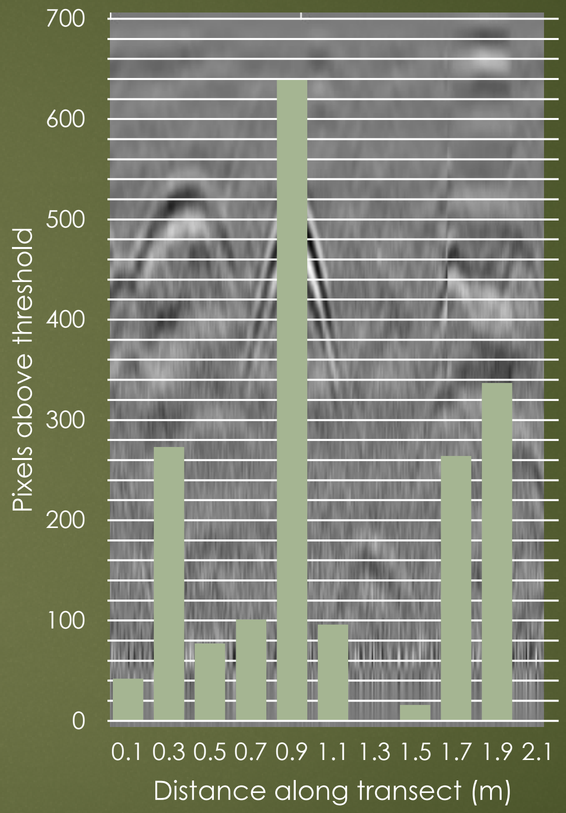

X	Y	Pixels
0.1	0	42
0.3	0	273
0.5	0	77
0.7	0	101
0.9	0	639
1.1	0	96
1.3	0	0
1.5	0	16
1.7	0	264
1.9	0	337
2.1	0	0

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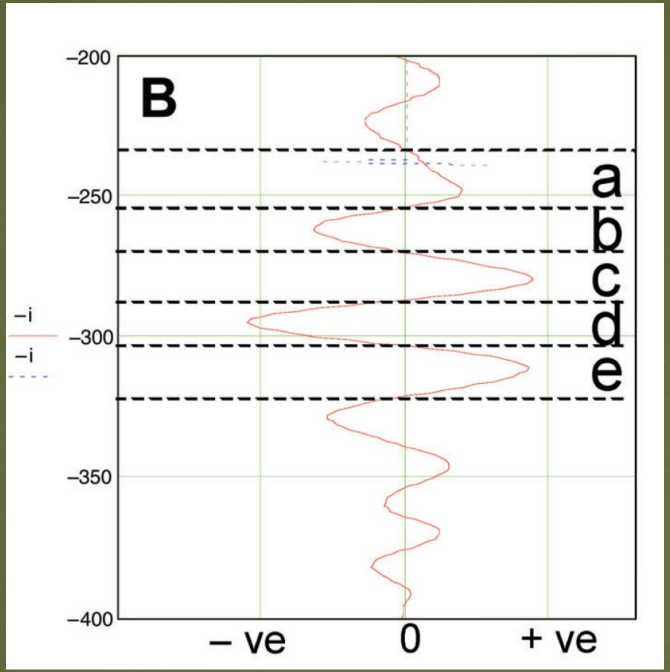
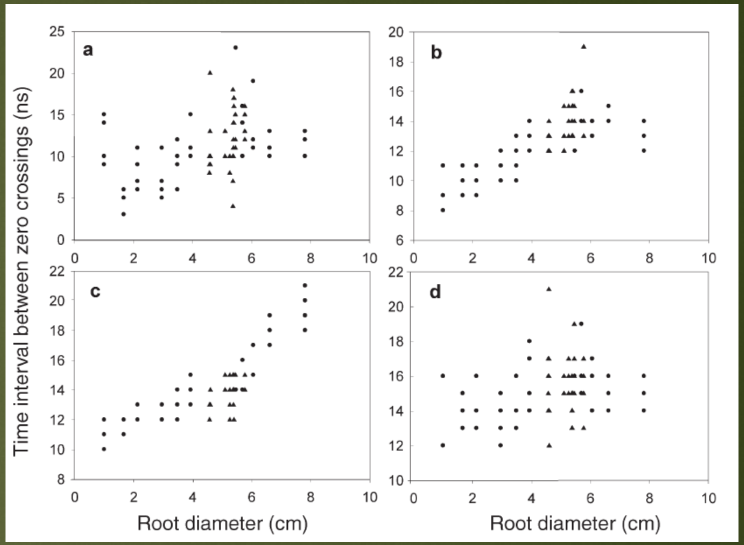
Can process and grid any number of image files automatically by X Y coordinates, ROI, and depth.

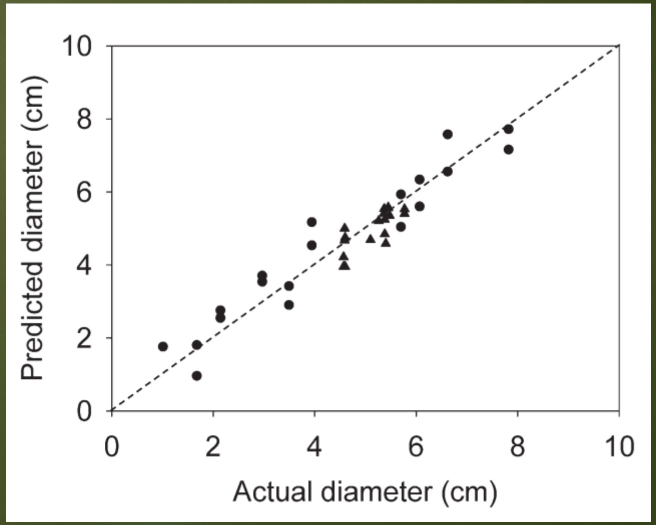
Approaches in waveform analysis

Waveform analysis

- Select waveforms manually or automatically
- Extract data from waveform
- High quality quantitative data





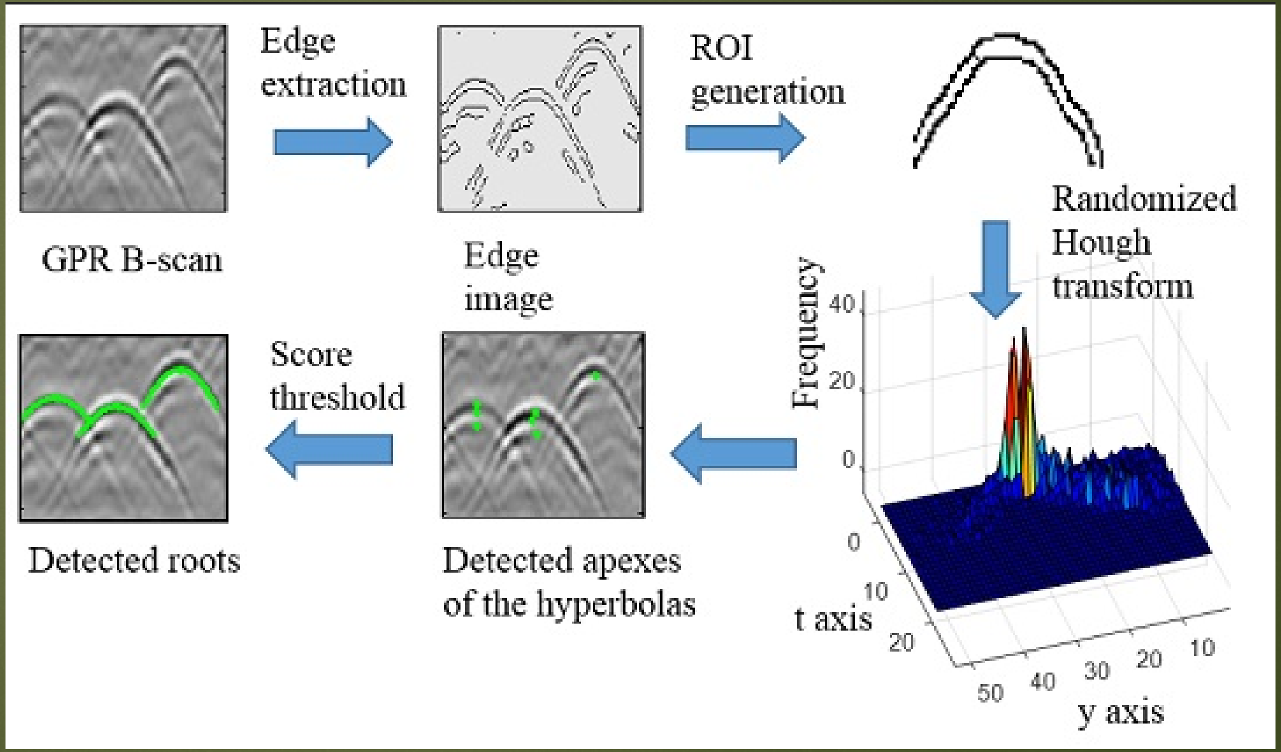


Barton and Montagu. 2004. Tree Physiology

Recent advances in waveform analysis

Waveform analysis

- Select waveforms manually or automatically
- Extract data from waveform
- High quality quantitative data

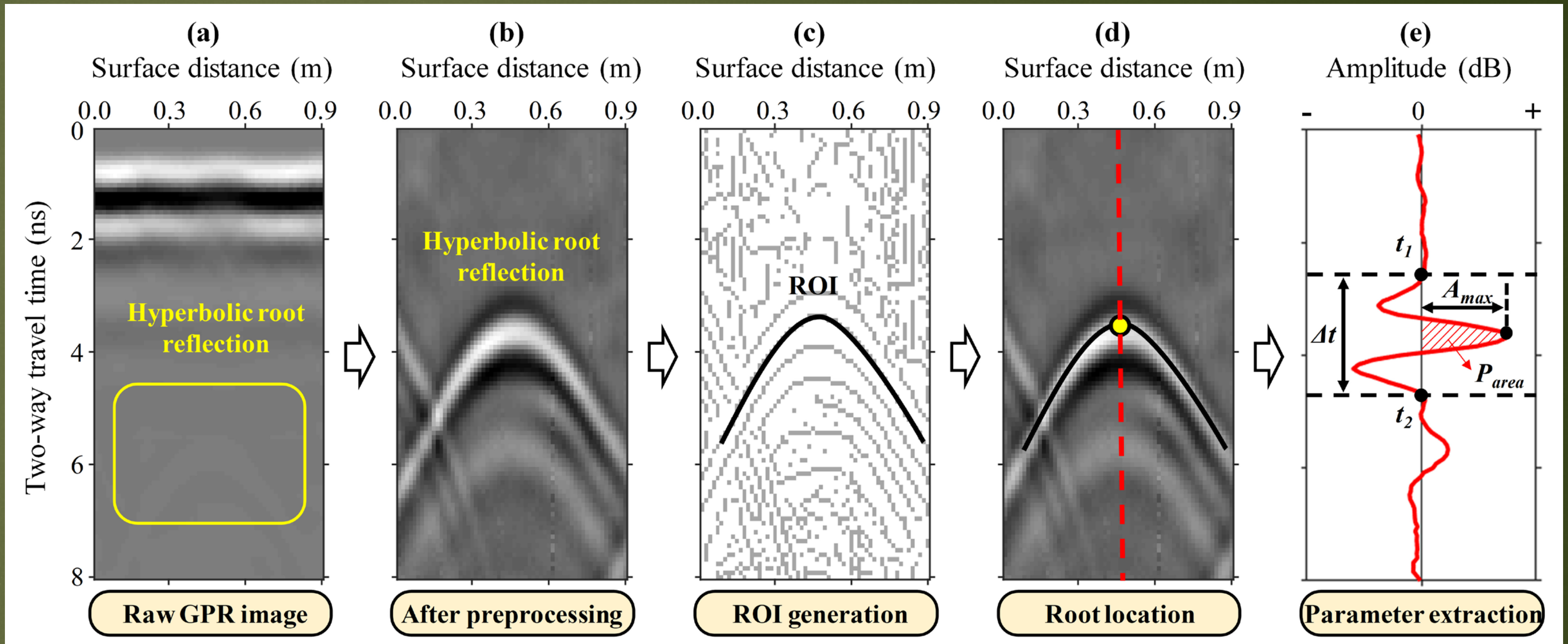


Li *et al.* 2016. Remote Sensing

Recent advances in waveform analysis

Waveform analysis

- Select automatically
- Extract data from waveform
- High quality quantitative data
  - Root detection and position (XYZ)
  - Root diameter
  - Rooting zone moisture content
  - Root water content



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Recent advances in root detection

Root angle and detection

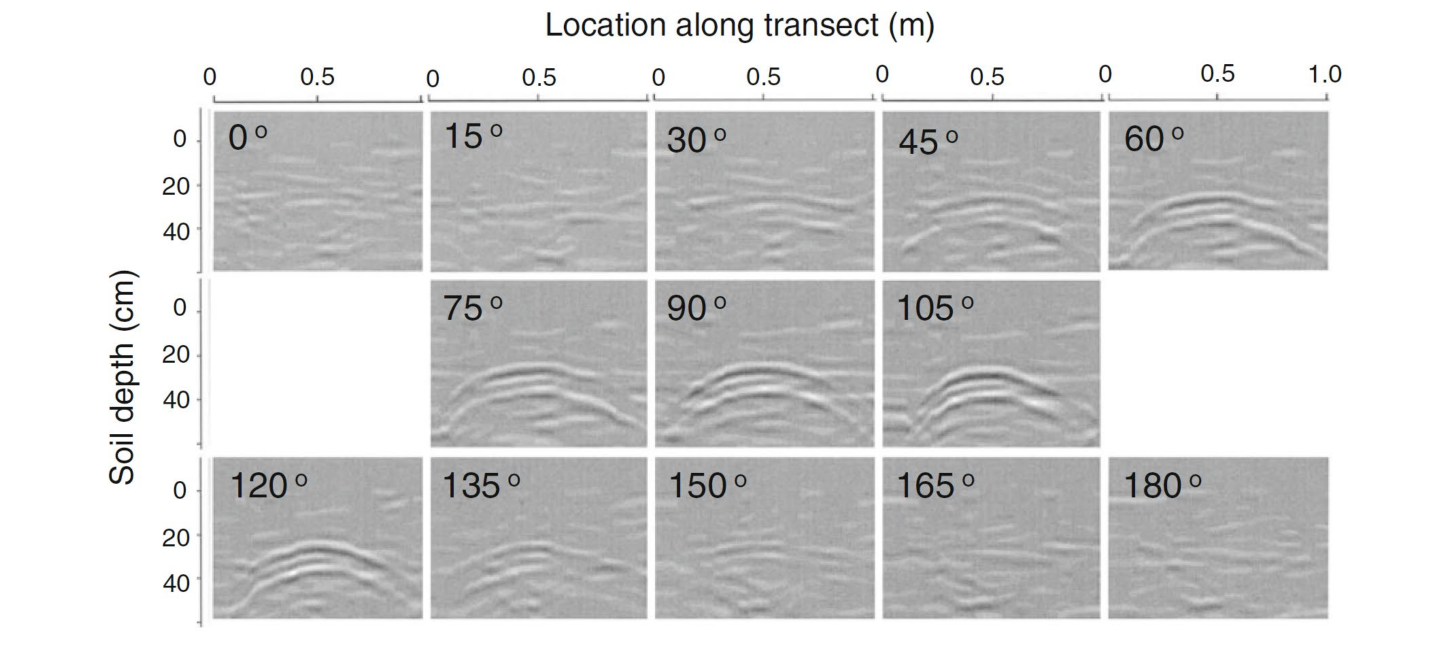
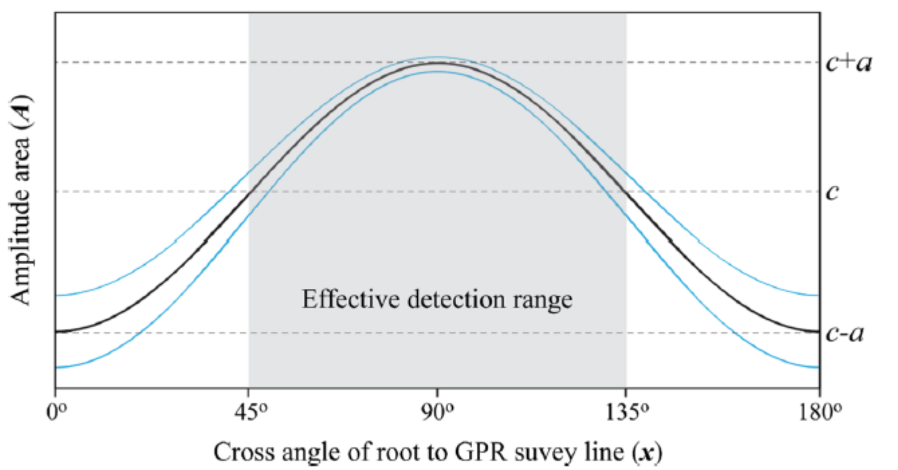


Fig. 3 Representative radar profiles of a buried 60-mm diameter dowel with angles changed in 15° steps. The horizontal axis represents the distance along the transect (m), and the vertical axis represents depth (cm)

Tanikawa *et al.* 2015. Plant and Soil

50 % of root angles are poorly detected in a linear transect



Guo *et al.* 2015. Plant and Soil

Recent advances in root detection

Role of image and waveform analysis

Image analysis

- Dependent on quality of b scan processing
- Sensitive to soil water
- Averages detectable and undetectable roots
- Excels at biomass

Waveform analysis

- Dependent on root angle
- Not sensitive to soil water
- Diverse quantitative variables

Recent advances in root detection

Questions?

Extra slides

What root and rhizosphere properties can be assessed?

- Biomass
- Spatial distribution
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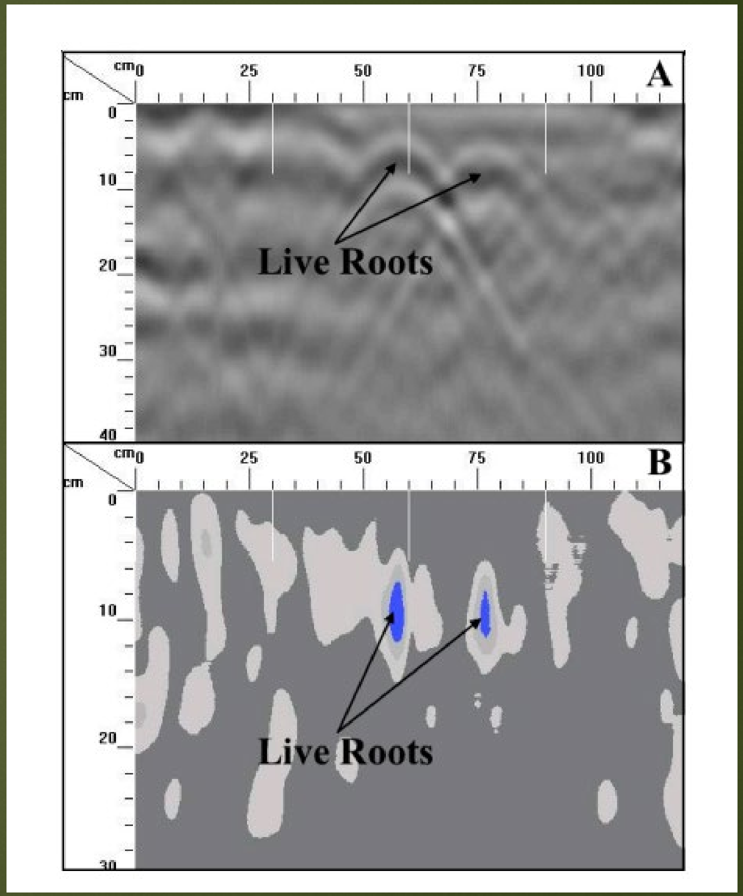

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